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Treehouse Design

user example

0001



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1.1 Summary

This report covers the structural design of a treehouse in Novato, California. The design follows the requirements of the California Building Code (CBC).

The treehouse is supported by a single Douglas fir beam spanning 12 feet between two trees. The beam supports a uniformly distributed load (UDL) from the treehouse floor and live load from occupants. The design includes calculations for the required beam size and material properties to ensure safety and compliance with building codes.

1.2 Load Combinations and Geometry

Dead and live load contributions to beam UDL.

Table 1: ASCE 7-05 Load Effects

Equation No.	Load Combination
16-1	$1.4(D+F)$
16-2	$1.2(D+F+T) + 1.6(L+H) + 0.5(L_r \text{ or } S \text{ or } R)$
16-3	$1.2(D+F+T) + 1.6(L_r \text{ or } S \text{ or } R) + (f_1 L \text{ or } 0.8W)$

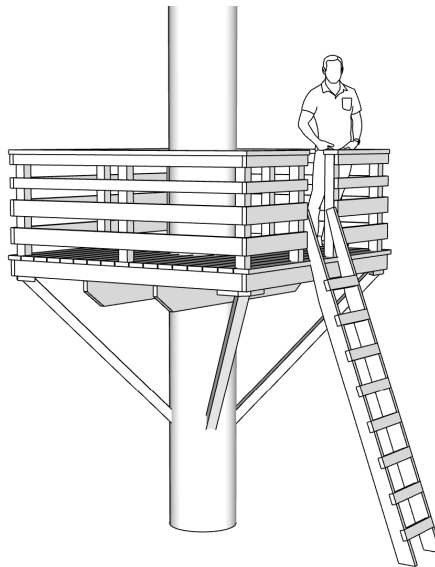


Fig. 1 - Treehouse

1.3 Symbols

Table 2: Math Symbols

Abbreviation	Definition
D	dead load
L	live load



Abbreviation	Definition
D_m	module dead load
E	earthquake load
F_a	acceleration site coefficient
F_v	velocity site coefficient
F_N	normal wind force
GC_M_s	net moment static coefficient
GC_{M_d}	net moment dynamic coefficient
GC_M	net moment coefficient
GC_P	net pressure coefficient
GC_P_s	net static pressure coefficient
GC_P_d	net dynamic pressure coefficient
k_1	hazard coefficient
k_2	terrain and structure coefficient
k_3	topography coefficient
Kzt	topographic Factor
K_z	velocity pressure exposure coefficient
MRI	mean return interval
p_d	net design wind pressure on module - Pa
SDOF	single degree of freedom
S_s	short period mapped acceleration
S_D_S	site design response acceleration
S_1	1 second period mapped acceleration
S_M_S	short period parameter
S_M_1	1 second period parameter
T	fundamental period of structure
M_t_o_r	wind moment about panel center
T_0	short period spectral cap
T_S	long period spectral cap
V_b	basic wind speed
V_B	seismic design base shear
W	wind load / seismic weight of structure

Table 3: Drawing Abbreviations

Abbreviation	Definition
ASD	Allowable Stress Design
ACI	American Concrete Institute



Abbreviation	Definition
AISC	American Institute of Steel Construction
AISI	American Iron and Steel Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
AB	Anchor Bolt
BDRY	Boundry
CBC	Califiornia Building Code
CRC	Califiornia Residential Code
CIP	Cast-In-Place
CLR	Clear
CONC	Concrete
CMU	Concrete Masonry Unit
CRSI	Concrete Reinforcing Steel Institute
CONST JT	Construction Joint
CONT	Continuous
CJ	Control Joint
D-C	Demand-Capacity (ratio)
DIA	Diameter
DIM	Dimension
EA	Each
EF	Each Face
EJ	Expansion Joint
ES	Each Side
EW	Each Way
EXP Bolt	Expansion Bolt
EXP JT	Expansion Joint
FTG	Footing
FND	Foundation
GALV	Galvanized
GA	Gauge
GR	Grade
HT	Height
IN	Inch
ID	Inside Diameter
ICBO	International Conference of Building Officials
K	Kip (1000 Pounds)



Abbreviation	Definition
LWC	Light Weight Concrete
LRFD	Load and Resistance Factor Design
NWC	Normal Weight Concrete
NIC	Not in Contract
OC	On Center
OD	Outside Diameter
OPNG	Opening
PVC	Polyvinyl Chloride
PSF	Pounds per Square Foot
PSI	Pounds per Square Inch
R	Radius
REINF	Reinforced
SIM	Similar
SOG	Slab on Grade
SL	Splice Length
SQ	Square
STD	Standard
SDI	Steel Deck Institute
SF	Step Footing or Square Foot
SYM	Symmetrical
THK	Thick or Thickness
T&B	Top and Bottom
T&G	Tongue and Groove
TOC	Top of Concrete
TOF	Top of Foundation
TOS	Top of Steel
TOW	Top of Wall
TYP	Typical
UNO	Unless Noted Otherwise
WWF	Welded Wire Fabric
W/	With
WP	Working Point

